improvement in the accuracy with which the positions, sizes, and shapes of the radio sources were measured.

Such an improvement was a tall order. An optical telescope, or even a naked human eye, can achieve a high resolution with ease, because the waves it works with (light) have very short wavelengths, less than 10^{-6} meter. By contrast, human ears cannot distinguish very accurately the direction from which a sound comes because sound waves have wavelengths that are long, roughly a meter. Similarly, radio waves, with their meter-sized wavelengths, give poor resolution—unless the telescope one uses is enormously larger than a meter. Reber's telescope was only modestly larger; hence, its modest resolution. To achieve a 100-fold improvement in resolution would require a telescope 100 times larger, roughly a kilometer in size, and/or the use of shorter wavelength radio waves, for example, a few centimeters instead of one meter.

The experimental physicists actually achieved this l00-fold improvement in 1949, not by brute force, but by cleverness. The key to their cleverness can be understood by analogy with something very simple and familiar. (This is just an analogy; it in fact is a slight cheat, but it gives an impression of the general idea.) We humans can see the three-dimensionality of the world around us using just two eyes, not more. The left eye sees around an object a little bit on the left side, and the right eye sees around it a bit on the right side. If we turn our heads over on their sides we can see around the top of the object a bit and around the bottom of the object a bit; and if we were to move our eyes farther apart (as in effect is done with the pair of cameras that make 3-D movies with exaggerated three-dimensionality), we would see somewhat farther around the object. However, our three-dimensional vision would not be improved enormously by having a huge number of eyes, covering the entire fronts of our faces. We would see things far more brightly with all those extra eyes (we would have a higher *sensitivity),* but we would gain only modestly in three-dimensional *resolution.*

Now a huge, 1-kilometer radio telescope (left half of Figure 9.2) would be somewhat like our face covered with eyes. The telescope would consist of a 1-kilometer-sized bowl covered with metal that reflects and focuses the radio waves up to a wire antenna and radio receiver. If we were to remove the metal everywhere except for a few spots widely scattered over the bowl, it would be like removing most of those extra eyes from our face, and